

5

Forks, Chains, and Immoralities

Welcome to Chapter 5!

In the previous chapter, we discussed the basic characteristics of graphs and showed how to use graphs to build graphical models. In this chapter, we will dive deeper into graphical models and discover their powerful features.

We'll start with a brief introduction to the mapping between distributions and graphs. Next, we'll learn about three basic graphical structures – forks, chains, and colliders – and their properties.

Finally, we'll use a simple linear example to show in practice how the graphical properties of a system can translate to its statistical properties.

The material discussed in this chapter will provide us with a solid foundation for understanding classic causal inference and *constraint-based* discovery methods and will prepare us for understanding other families of algorithms that we'll introduce in *Parts 2* and *3* of this book.

In this chapter, we cover the following key topics:

- Graphs and distributions and how to map between them
- Forks, chains, and colliders or...immoralities
- Forks, chains, colliders, and regression

Graphs and distributions and how to map between them

In this section, we will focus on the mappings between the statistical and graphical properties of a system.

To be more precise, we'll be interested in understanding how to translate between graphical and statistical independencies. In a perfect world, we'd like to be able to do it in both directions: from graph independence to statistical independence and the other way around.

It turns out that this is possible under certain assumptions.

The key concept in this chapter is one of *independence*. Let's start by reviewing what it means.

How to talk about independence

Generally speaking, we say that two variables, X and Y , are independent when our knowledge about X does not change our knowledge about Y (and vice versa). In terms of probability distributions, we can express it in the following way:

$$P(Y) = P(Y|X)$$

$$P(X) = P(X|Y)$$

In other words: the marginal probability of Y is the same as the conditional probability of Y given X and – respectively – the marginal probability of X is the same as the conditional probability of X given Y .

In plain English: learning something new about X will not change our beliefs about Y (and the other way around).

Probabilities and beliefs

You might have noticed that we talk about “knowledge” and “beliefs” in the context of probability here. This vocabulary is usually associated with the Bayesian approach to probability (as opposed to the frequentist approach). We use the Bayesian approach here because we believe it brings a more intuitive perspective on independence. At the same time, we use both frequentist and Bayesian concepts in this book (and discuss them where we believe it’s important). Bayesian and frequentist are two different approaches to statistics. They converge to virtually identical results in large sample sizes but put accents slightly differently (e.g., we can sensibly talk about *beliefs* in the Bayesian framework but not really in the frequentist framework).

For an introduction to some of the fundamental differences between the two approaches, check out Jake VanderPlas’ talk (<https://bit.ly/BayesianVsFrequentist>). For slightly more formal treatment, check out the **Massachusetts Institute of Technology (MIT)** course notes PDF (<https://bit.ly/BayesVsFreqMIT>).

One interesting consequence of the independence between two variables is that their joint distribution factorizes into the product of the marginals (Bishop, 2006):

$$P(X, Y) = P(X)P(Y)$$

More general notation for independence involves the symbol, $\perp$ (usually called *double up tack*), whose form visually encodes the notion of orthogonality (<https://bit.ly/OrthogonalityMath>). Using $\perp$, we can express the fact that X and Y are independent in the following way:

$$X \perp Y$$

The concept of independence plays a vital role in statistics and causality. As we will learn soon, its generalization – **conditional independence** – is even more important. We say that X and Y are conditionally independent given Z , when X does not give us any new information about Y assuming that we observed Z .

Let's see how to encode *conditional independence* using the notation we just introduced.

We can express the fact that X is independent of Y given Z in the following way:

$$X \perp\!\!\!\perp Y|Z$$

Equivalently, in terms of probabilities:

$$P(X, Y|Z) = P(X|Z)P(Y|Z)$$

Note that we factorized the joint distribution of X and Y (given Z) into a product of two simple conditionals ($X|Z$ and $Y|Z$) using the property introduced earlier ($P(X, Y) = P(X)P(Y)$).

Now, let's introduce a simple yet useful distinction. We will use the symbol, $\perp_p$, to denote *independence in the distribution* and $\perp_G$ to denote *independence in the graph*. To say that X and Y are independent in their distributions, we'll use the following:

$$X \perp_p Y$$

On the other hand, to say that X and Y are independent in the graph, we'll say:

$$X \perp_G Y$$

Equipped with this new shiny notation, let's discuss why mapping between graphical and distributional independence might be important to us.

Choosing the right direction

The basic goal of *causal inference* is to estimate the causal effect of one set of variables on another. In most cases, to do it accurately, we need to know which variables we should control for. We've seen in previous chapters that to accurately control for confounders, we need to go beyond the realm of pure statistics and use the information about the data-generating process, which can be encoded as a (causal) graph. In this sense, the ability to translate between graphical and statistical properties is central to causal inference.

Having the translation ability in the other direction – from statistical to graphical properties – is essential for *causal discovery*, a process that aims at recreating the causal graph from observational and/or interventional data.

To enable the mappings in both directions, we need to meet certain criteria. Let's revise them.

Independence in a graph – a working definition

We say that two nodes are *unconditionally* (or marginally) *independent* in the graph when there's *no open path* that connects them *directly* or *indirectly*.

We say that two nodes, X and Y , are *conditionally independent* given (a set of) node(s) Z when Z blocks *all open paths* that connect X and Y .

In other words, (in)dependence in the graph is a function of open paths between nodes in this graph. We'll use and enrich this intuition in the current chapter and build a more formal perspective on top of it in *Chapter 6*.

Conditions and assumptions

We'll start the discussion with assumptions that are important from the causal *inference* point of view. After this, we'll move to assumptions related to causal *discovery*, and we'll finish with a brief overview of a more general concept of causal sufficiency.

Conditions for causal inference

From the *causal inference* point of view, we need to make sure that we can map the graphical (conditional) independencies into statistical (conditional) independencies.

In order to achieve this we need to satisfy the **causal Markov condition** (also known as the *causal Markov assumption* or *(local) Markov property*).

The *causal Markov condition* states that the node, V_i , is independent of all its non-descendants (excluding its parents) given its parents. Therefore, formally, it can be presented as follows:

$$V_i \perp\!\!\!\perp_G V_j \mid PA(V_i) \quad \forall_{j \in G(V, E) \setminus \{DE(V_i), PA(V_i)\}}$$

This formula might look discouraging, but in fact, it says something relatively simple. Let's dissect and analyze it step by step:

- We start with $V_i \perp\!\!\!\perp_G V_j$. The symbol, $\perp\!\!\!\perp_G$, denotes independence in the graph. Therefore, the statement says that nodes, V_i and V_j , are independent in the graph or – in other words – that there are no open paths between them.

Next, we have the conditioning bar, $\mid$, and $PA(V_i)$. The latter stands for *parents of node V_i* . These are all the nodes that have edges *directed at* V_i . This conditioning part of the formula informs us that the independence between V_i and V_j only holds when we control for (the set of) parents of the node, V_i .

Next, we have $\forall_{j \in G(V, E)}$. The $\forall$ symbol means “*for all*”. What we say here is that the node, V_i , is independent of all other nodes ($j \neq i$) in the graph $G(V, E)$, where G represents our graph (**directed acyclic graph (DAG)**) of interest, V is a set of all vertices (nodes) in this graph, and E is a set of all edges in this graph.

Finally, we have the cryptic $\setminus \{DE(V_i), PA(V_i)\}$. Let's start with the $\setminus$ symbol. We read it as “excluding”. Curly brackets denote a set, and we have two elements in this set:

- $DE(V_i)$, which represents a set of all *descendants* of the node, V_i
- $PA(V_i)$, which represents a set of *parents* of V_i

Putting it all together: the node, V_i , is independent of all other nodes in the graph, G , excluding the *descendants* and *parents* of this node, given its *parents*. If this is not entirely clear to you at this stage, that's perfectly fine. We'll reiterate this concept while discussing **d-separation** in the next chapter.

In the meantime, feel free to take a look at *Figure 5.1* and *Figure 5.2*, which depict two examples of graphs in which the causal Markov condition holds. In both cases controlling for $PA(V_i)$ – the parent node of the node, V_i , removes the association between nodes V_i and V_j . Note that the relationships between V_i and V_j are different in both figures. In *Figure 5.1*, the nodes, V_i and V_j , have a common cause ($PA(V_i)$), in *Figure 5.2*, the node, V_j is a grandparent of V_i . If the causal Markov condition holds, the association between the nodes V_i and V_j should be removed in both cases. Note that if there was an unobserved common cause of both nodes (V_i and V_j) in any of the scenarios, controlling for $PA(V_i)$ would not render the nodes independent, and the condition would be violated:

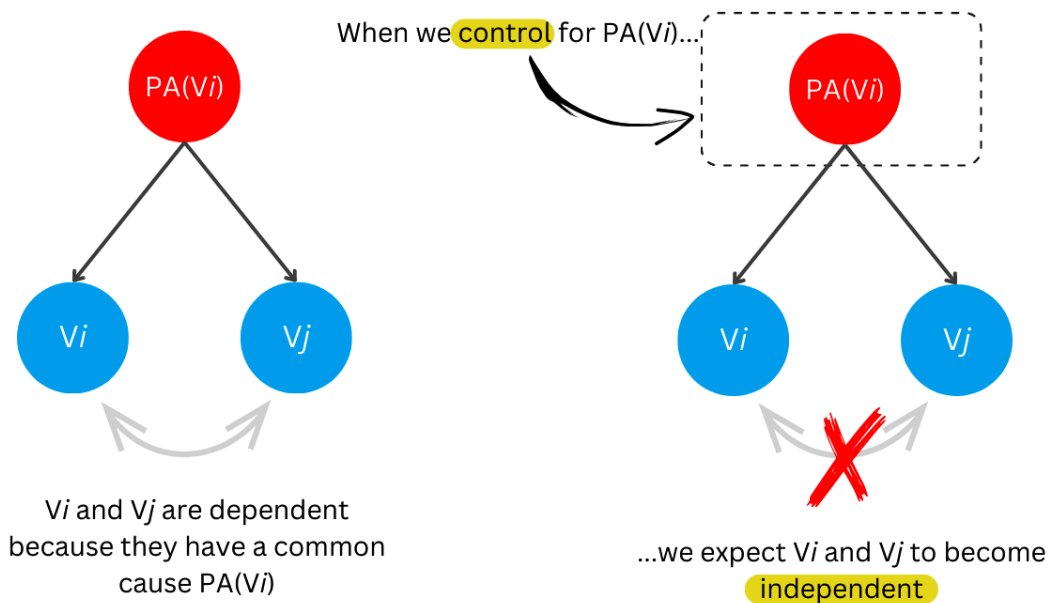


Figure 5.1 – Causal Markov condition – example one

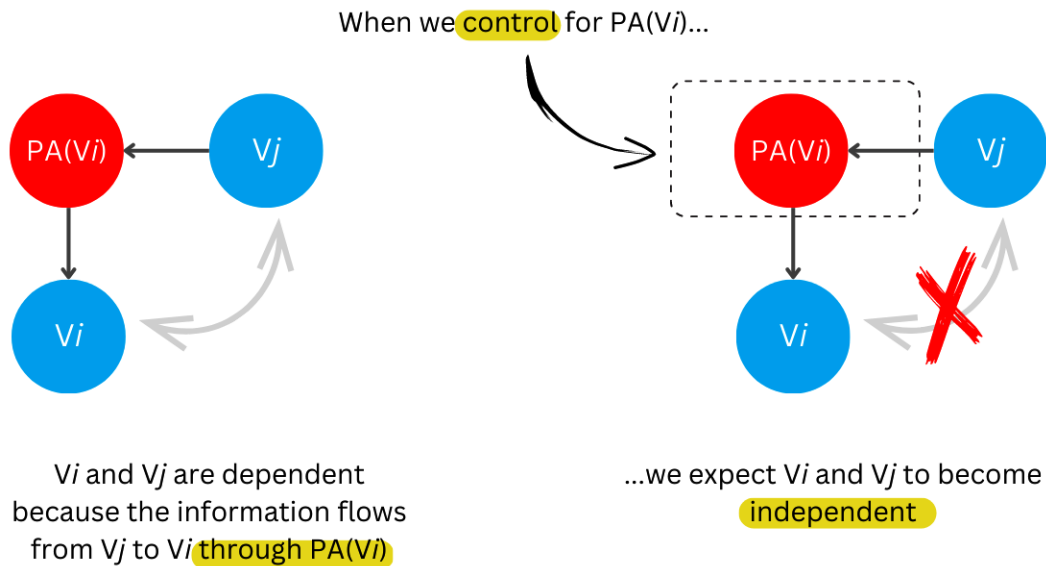


Figure 5.2 – Causal Markov condition – example two

The important statement is that when the causal Markov condition holds, the following is true:

$$X \perp\!\!\!\perp_G Y|Z \Rightarrow X \perp\!\!\!\perp_P Y|Z$$

We read it as: X and Y are independent in the graph given Z , then they are also statistically independent given Z . This is known as the **global Markov property**. In fact, we could show that the *global Markov property*, the *local Markov property*, and another property called the *Markov factorization property* are equivalent (Lauritzen 1996, pp. 51-52 for a formal proof; Peters et al., 2017, pp. 101 for an overview).

Assumptions for causal discovery

So far, we have discussed the importance of mapping between graphical and distributional independence structures. Now, let's reverse the direction. Causal discovery aims at discovering (or learning) the true causal graph from observational and/or (partially) interventional data. In general, this task is difficult. Nonetheless, it's possible when certain conditions are met.

Causal discovery has a couple of different flavors. In this section, we'll focus on a family of methods called **constrain-based causal discovery** (sometimes also called **independence-based causal discovery**). These methods are designed to find statistical independencies in the observational data and try to recreate the true causal graph from these independencies. One of the main assumptions behind this family of methods is called the **faithfulness assumption**. Its simplest formulation is the following:

$$X \perp\!\!\!\perp_P Y|Z \Rightarrow X \perp\!\!\!\perp_G Y|Z$$

This might look familiar. Note that it's the exact reverse of the global Markov property that we discussed in the previous section. The formula says that if X and Y are independent in the distribution given Z , they will also be independent in the graph given Z .

As we mentioned in the previous chapter, the faithfulness assumption might be difficult to fulfill sometimes. The most critical reason for this is the estimation error when testing for conditional independence in the finite sample size regime (Uhler et al., 2013). Moreover, it's not very difficult to find situations where the assumption is violated. Intuitively, any situation where one variable influences another through two different paths and these paths cancel each other out completely would lead to the violation of *faithfulness*. That said, although it's relatively easy to come up with such examples (for example, Neal, 2020, pp. 100-101; Peters et al., 2017, pp. 107-108), the probability of encountering them in the real world is very small (Sprites et al., 2000, pp. 68-69). Therefore, the first problem is much more serious and more difficult to tackle in practice than the second one.

The last assumption that we will discuss in this section is called the **causal minimality condition** (also known as the **causal minimality assumption**).

Although the *global Markov property* that we discussed earlier is pretty powerful, it leaves us with some uncertainty regarding the true structure of a graph we're trying to retrieve.

It turns out that there might be more than one graph that entails the same distribution! That's problematic when we want to recover causal structure (represented as a causal graph) because the mapping between the graph and the distribution is ambiguous. To address this issue, we use the causal minimality condition.

The causal minimality assumption states that DAG G is minimal to distribution, P , if and only if G induces P , but no proper sub-graph of G induces P . In other words, if graph G induces P , removing *any* edge from G should result in a distribution that is different than P .

Causal minimality can be seen from various perspectives (Neal, 2020, pp. 21-22; Pearl, 2009, p. 46; Peters & Schölkopf, 2017, pp. 107-108). Although the assumption is usually perceived as a form of *Ockham's razor*, its implications have practical significance for constraint-based causal discovery methods and their ability to recover correct causal structures.

Other assumptions

Before we conclude this section, let's discuss one more important assumption that is very commonly used in causal discovery and causal inference. It's an assumption of **no hidden confounding** (sometimes also referred to as **causal sufficiency**). Although meeting this assumption is not necessary for *all* causal methods, it's pretty common.

Note that causal sufficiency and the causal Markov condition are related (and have some practical overlap), but they are not identical. For further details, check out Scheines (1996).

In many real-world scenarios, we might find it challenging to verify whether hidden confounding exists in a system of interest.

In this section, we reviewed the concept of statistical and graphical independence, discussed the motivations for mapping between graphical and statistical independencies, and examined conditions that allow us to perform such mappings.

Now, let's discuss three basic graphical structures that are immensely helpful in determining sets of conditional independencies.

Ready?

Chains, forks, and colliders or...immoralities

On a sunny morning of June 1, 2020, Mr. Huang was driving his gleaming white Tesla on one of Taiwan's main highways. The day was clear, and the trip was going smoothly. Mr. Huang engaged the autopilot and set the speed to 110 km/h. While approaching the road's 268-kilometer mark, he was completely unaware that only 300 meters ahead, something unexpected was awaiting him. Nine minutes earlier, another driver, Mr. Yeh, had lost control of his vehicle. His white truck was now overturned, almost fully blocking two lanes of the highway right at the 268.3-kilometer mark.

Around 11 seconds later, to Mr. Huang's dismay, his Tesla crashed into the overturned truck's rooftop. Fortunately, the driver, Mr. Huang, survived the crash and came out of the accident without any serious injuries (Everington, 2020).

A chain of events

Many modern cars are equipped with some sort of collision warning or collision prevention system. At a high level, a system like this consists of a detector (or detector module) and an alerting system (sometimes also an automatic driving assistance system). When there's an obstacle on the collision course detected by the detector, it sends a signal that activates the alerting system. Let's say that the detector is in state 1 when it detects an obstacle and it's in state 0 when it detects no obstacles.

Figure 5.3 presents a DAG representing our collision warning system:

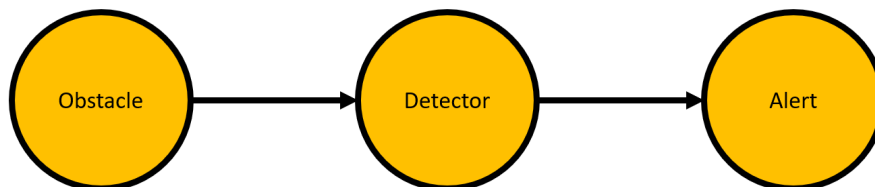


Figure 5.3 – A DAG representing a collision warning system

An important fact about a system such as the one presented in *Figure 5.3* is that the existence of the obstacle does not give us any new information about the alert when we know the detector state. The obstacle and the alert are *independent, given the detector state*.

In other words, when the detector *thinks* there is a pedestrian in front of the car, the alarm will set off even if there's no pedestrian in front of the car. Similarly, if there's an obstacle on the road and the detector does not recognize it as an obstacle, the alert will remain silent.

In particular, if a detector is based on a computer vision system and the model *hallucinates* a pedestrian on the car's course, the alert will set off. On the other hand, when there's a real pedestrian in front of the car and the model fails to recognize them, the alert will remain silent.

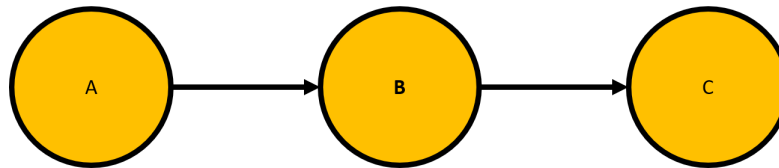
An example of an unexpected detector system behavior went viral in August 2022. A TikTok, RealRusty, shared a video showing how his car's object detection system reacted to a horse-drawn carriage (<https://bit.ly/HorseVsTesla>).

Another great example of a chain of causally-linked events comes from Judea Pearl's and Dana Mackenzie's *The Book of Why* (Pearl & Mackenzie, 2019, pp. 113-114). The authors discuss the mechanics of a fire alarm. The chain is *fire* $\rightarrow$ *smoke* $\rightarrow$ *alarm* because the alarm is actually controlled by a smoke particle detector rather than any other indicator of fire such as temperature.

Let's abstract the structure that these two examples share.

Chains

First, let's replace the variables in the graph from *Figure 5.3* with *A*, *B*, and *C*. This general structure is called a **chain**, and you can see it in *Figure 5.4*:



Chain 

Figure 5.4 – A chain structure

We can generalize the independence property from our collision warning example to *all* chain structures. It means that in every structure that can be encoded as $A \rightarrow B \rightarrow C$, *A* and *C* are independent in the graph given *B*. Therefore, formally, it can be presented as follows:

$$A \perp\!\!\!\perp_C C | B$$

Intuitively, controlling for *B* closes the only open path that exists between *A* and *C*. Note that *A* and *C* become *dependent* when we *do not* control for *B*.

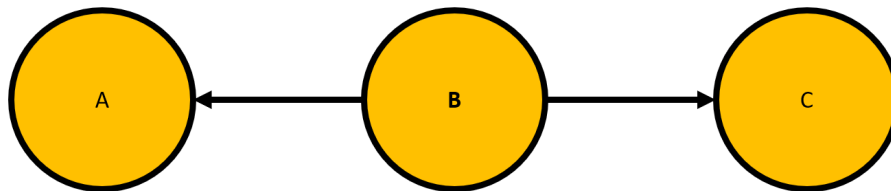
Translating this back to our object detection system: if we do not observe the detector state, the presence of the obstacle within the system's range becomes correlated with the safety response (alarm and emergency breaking).

You might already see where it's going. If we're able to fulfill the assumptions that we discussed in the previous section (causal Markov condition, no hidden confounding), we can now predict conditional independence structure in the data from the graph structure itself (and if that's true we can also figure out which variables we should control for in our model to obtain valid causal effect estimates(!) – we'll learn more on this in the next chapter). Moreover, predicting the graph structure from the observational data alone also becomes an option. That's an exciting possibility, but let's take it slowly.

Now, let's see what happens if we change the direction of one of the edges in our *chain* structure.

Forks

Figure 5.5 represents a **fork**. A fork is a structure where the edge between nodes *A* and *B* is reversed compared to the *chain* structure:



Fork 🍴

Figure 5.5 – A fork structure

In the fork, node *B* becomes what we usually call a **common cause** of nodes *A* and *C*.

Imagine you're driving your car, and you suddenly see a llama in the middle of the road.

The detector recognizes the llama as an obstacle. The emergency brake kicks in before you even realize it. At the same time, a small subcortical part of your brain called the *amygdala* sends a signal to another structure, the *hypothalamus*, which in turn activates your *adrenal glands*. This results in an *adrenaline* injection into your bloodstream. Before you even notice, your body has entered a state that is popularly referred to as a *fight-or-flight* mode.

The presence of a llama on the road caused the detector to activate the emergency brake and caused *you* to develop a rapid response to this potentially threatening situation.

This makes the llama on the road a common cause of your stress response and the detector's response.

However, when we control for the llama on the road, your threat response and the detector's response become independent. You might feel stressed because you're running late for an important meeting or because you had a tough conversation with your friend, and this has zero connection to the detector state.

Note that in the real world, you would also likely react with a *fight-or-flight* response to the mere fact that the emergency brakes were activated. The path, *llama* $\rightarrow$ *detector* $\rightarrow$ *brakes* $\rightarrow$ *fight-or-flight*, will introduce a spurious connection between *llama* and *fight-or-flight* that can be removed by controlling for the *brakes* variable.

Let's build a more formal example to clear any confusion that can come from this connection.

Let's take a look at *Figure 5.5* once again and think about conditional independencies. Are *A* and *C* *unconditionally dependent*?

In other words: are *A* and *C* dependent when we *do not* control for *B*? The answer is *yes*.

Why are they dependent? They are dependent because they both *inherit* some information from *B*. At the same time, the information *inherited* from *B* is all they have in common.

Let's take a look at a simple structural model that describes a fork structure to see how independence manifests itself in the distribution:

$$U_A \sim \mathcal{N}(0, 1)$$

$$U_B \sim \mathcal{N}(0, 1)$$

$$U_C \sim \mathcal{N}(0, 1)$$

$$B := U_B$$

$$A := B + U_A$$

$$C := B + U_C$$

In the preceding formula, $U_{X \in \{A, B, C\}}$ represents independently distributed noise variables (here, they follow a normal distribution, but they don't have to in general) and $:=$ is the assignment operator that you might remember from previous chapters.

Now imagine that we only look at observations where $B = 0$.

What would happen?

When $B = 0$, then *A* and *C* will only be influenced by their respective noise terms that are independent by definition.

We can therefore conclude that in forks, *A* and *C* are conditionally independent given *B*. Formally, it can be presented as follows:

$$A \perp\!\!\!\perp C | B$$

In case we do not control for B , A and C are *dependent*. This independence pattern is identical to the one that we've obtained from the *chain* structure.

This is not great news.

It seems that chains and forks lead to the same pattern of conditional independence, and if we want to recreate a true graph from the data, we end up not knowing how to orient the edges in the graph!

This might sound disappointing, but before we let the disappointment take over, let's examine the last structure in this section.

Colliders, immoralities, or v-structures

As you can see, the **collider** has many names. Is that the only thing that makes this structure special? Let's check!

Let's take a look at *Figure 5.6*. In the *collider*, causal influence flows from two different parents into a single child node:

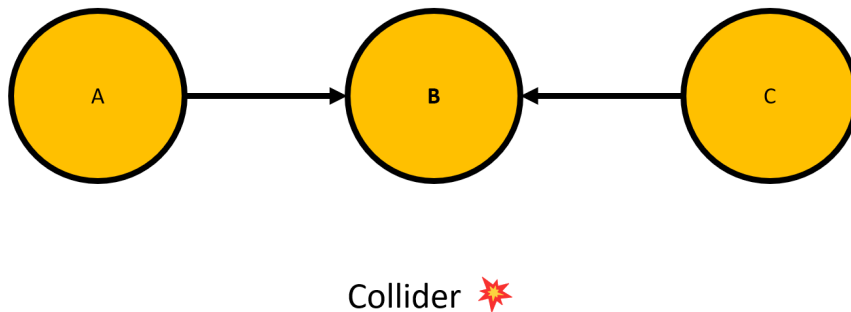


Figure 5.6 – A collider structure

The most important characteristic of a collider is that its independence pattern is *reversed* compared to chains and forks. So, what does it mean?

In colliders, A and C are *unconditionally independent*. Formally speaking, this can be represented as follows:

$A \perp\!\!\!\perp C$

When we control for B , they become *dependent*! Formally represented as follows:

$A \not\perp\!\!\!\perp C | B$

In the preceding formula, the slashed $\perp$ symbol reads *is dependent*. As you can see, this pattern is exactly the reverse of the one that we've seen for chains and forks!

And this time that's great news!

We can leverage the collider structure to unambiguously orient edges in a graph every time we see it (there might be some challenges to this, but we skip them for now). Moreover, if we're lucky, colliders can help us orient ambiguous edges coming from chains and forks. We'll see an example of such behavior in *Part 3, Causal Discovery*.

To extend our series of driving examples, let's think about llamas on the road and driving against the sun. I think that most people would agree it's reasonable to say that whether you drive against the sun is independent of whether there's a llama on the road.

Now, let's assume that your detector reacts with some probability to the sun reflexes as if there was an obstacle on the road.

In this case, whether there's a llama on the road becomes (negatively) correlated with whether you drive against the sun if you only control for the detector reaction.

Figure 5.7 symbolically demonstrates the loss of independence between the two when we control for the detector state:

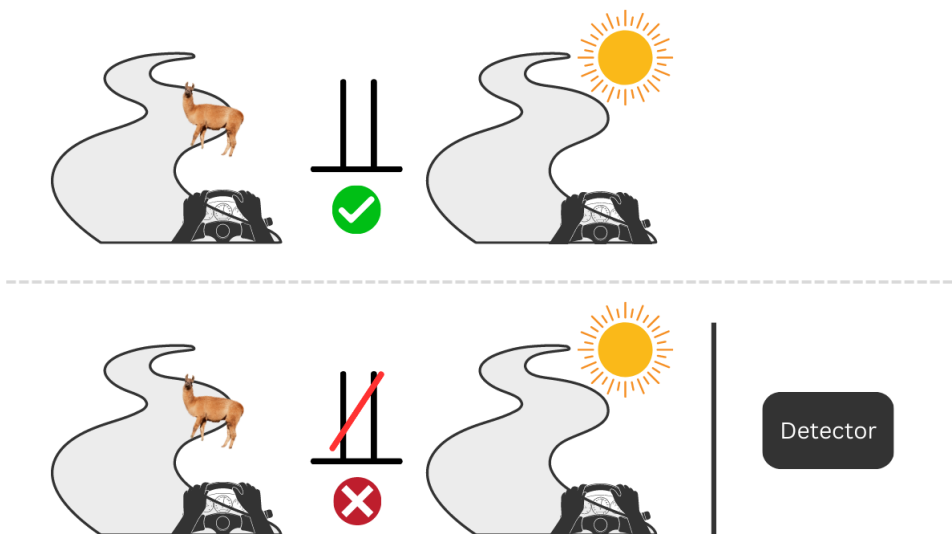


Figure 5.7 – A visual representation of independence structures in the collider example

The loss of independence between two variables, when we control for the collider, sounds surprising to many people at first. It was also my experience.

I found that for many people, examples involving real-world objects (even if these objects are llamas) bring more confusion than clarity when it comes to colliders.

Let's build a slightly more abstract yet very simple example to make sure that we clarify any confusion you might still have.

Take a look at *Figure 5.6* once again to refresh the graphical representation of the problem.

Now, let's imagine that both A and C randomly generate integers between 1 and 3. Let's also say that B is a sum of A and C . Now, let's take a look at values of A and C when the value of $B = 4$. The following are the combinations of A and C that lead to $B = 4$:

- $A = 1, C = 3$
- $A = 2, C = 2$
- $A = 3, C = 1$

Can you see the pattern? Although A and C are unconditionally independent (there's no correlation between them as they randomly and independently generate integers), they become correlated when we observe B ! The reason for this is that when we hold B constant and the value of A increases, the value of C has to decrease if we want to keep the value of B constant.

If you want to make it more visual, you can think about two identical glasses of water. If you randomly pour some water from one glass to the other, the total amount of water in both glasses will remain the same (assuming that we don't spill anything). If you repeat this n times and measure the amount of water in both glasses at each stage, the amount of water between the glasses will become negatively correlated.

I hope that this example will help you cement your intuition about colliders' conditional independence properties.

Thanks to the unique properties of colliders, they can be immensely helpful when we're trying to recover graph structures from observational data. Moreover, we can sometimes use colliders to disambiguate neighboring structures (we'll see an example of this in *Part 3, Causal Discovery*).

Unfortunately, this is not always the case. Let's discuss these scenarios now.

Ambiguous cases

As we've seen earlier, various graphical configurations might lead to the same statistical independence structure. In some cases, we might get lucky and have enough colliders in the graph to make up for it. In reality, though, we might often not be that fortunate.

Does this mean that the discussion we have had so far leads us to the conclusion that, in many cases, we simply cannot recover the graph from the data?

That's not entirely true. Even in cases where some edges cannot be oriented using constraint-based methods, we can still obtain some useful information!

Let's introduce the concept of the **Markov equivalence class (MEC)**. A set of DAGs, $\mathcal{D} = \{G_0(V, E_0), \dots, G_n(V, E_n)\}$, is *Markov equivalent* if and only if all DAGs in $\mathcal{D}$ have the same skeleton and the same set of colliders (Verma & Pearl, 1991).

A **skeleton** is basically an undirected version of a DAG – all the edges are in place, but we have no information on the arrows. If we add the edges for all the collider structures that we’ve found, we will obtain a **complete partially-directed acyclic graph (CPDAG)**.

If we take the CPDAG and generate a set of all possible DAGs from it, we’ll obtain a *MEC*. MECs can be pretty useful. Even if we cannot recover a full DAG, a MEC can significantly reduce our uncertainty about the causal structure for a given dataset.

Before we conclude this section, let’s take a look at *Figure 5.8*, which presents a simple MEC:

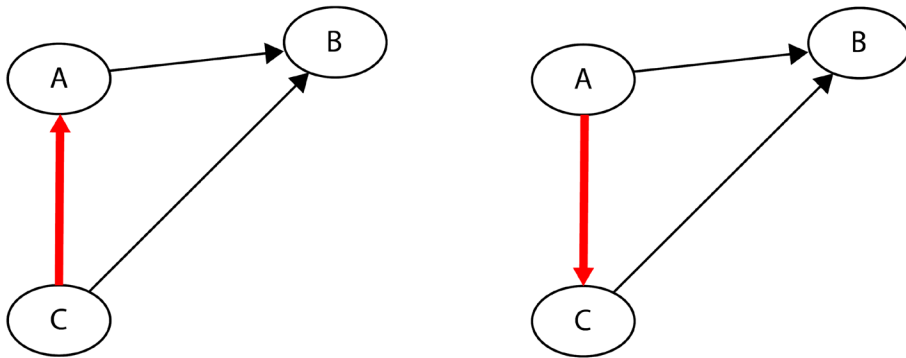


Figure 5.8 – An example of a MEC

The graphs in *Figure 5.8* have the same set of edges. If we removed the arrows and left the edges undirected, we would obtain two *identical* graphs, which is an indicator that both graphs have the same *skeleton*.

The collider ($A \rightarrow B \leftarrow C$) is present in both graphs. The only difference between the two graphs is the direction of the edge between nodes A and C. We can conclude that – consistently with our definition – the graphs in *Figure 5.8* meet the criteria for constituting a MEC.

Great!

Now, having a solid understanding of the three basic conditional independence structures – *chains*, *forks*, and *colliders* – we’re ready to put this knowledge into action!

Forks, chains, colliders, and regression

In this section, we will see how the properties of chains, forks, and colliders manifest themselves in regression analysis. The very type of analysis that we’ll conduct in this section is actually at the heart of some of the most classic methods of causal inference and causal discovery that we’ll be working with in the next two parts of this book.

What we're going to do now is to generate three datasets, each with three variables, *A*, *B*, and *C*. Each dataset will be based on a graph representing one of the three structures: a chain, a fork, or a collider. Next, we'll fit one regression model per dataset, regressing *C* on the remaining two variables, and analyze the results. On the way, we'll plot pairwise scatterplots for each dataset to strengthen our intuitive understanding of a link between graphical structures, statistical models, and visual data representations.

Let's start with graphs. *Figure 5.9* presents chain, fork, and collider structures:

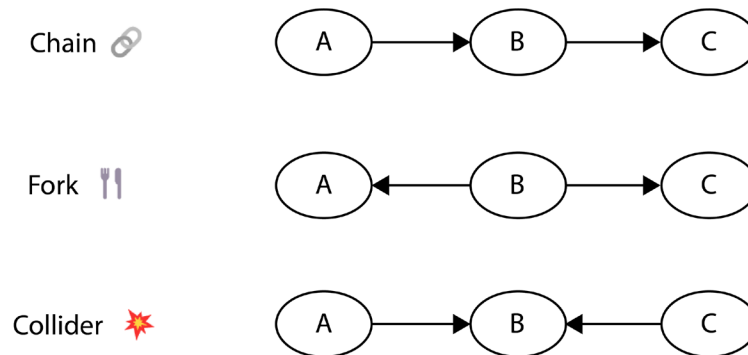


Figure 5.9 – Graphical representations of chain, fork, and collider structures

We will use the graphs from *Figure 5.9* to guide our data-generating process. Note that we omitted the *noise variables* for clarity of presentation.

The code for this section can be found in the `Chapter_05.ipynb` notebook (<https://bit.ly/causal-ntbk-05>).

First, let's define the general parameters:

```
NOISE_LEVEL = .2  
N_SAMPLES = 1000
```

`NOISE_LEVEL` will determine the standard deviation of noise variables in our datasets. `N_SAMPLES` simply determines the sample size.

Generating the chain dataset

Now, we're ready to generate the data for the chain structure:

```
a = np.random.randn(N_SAMPLES)
b = a + NOISE_LEVEL*np.random.randn(N_SAMPLES)
c = b + NOISE_LEVEL*np.random.randn(N_SAMPLES)
```

Our code follows the logic of a chain-structured graph – A directly influences B (but not C) and B influences C (but not A). C does not have any further influence.

Let's plot pairwise scatterplots for this dataset. We present them in *Figure 5.10*:

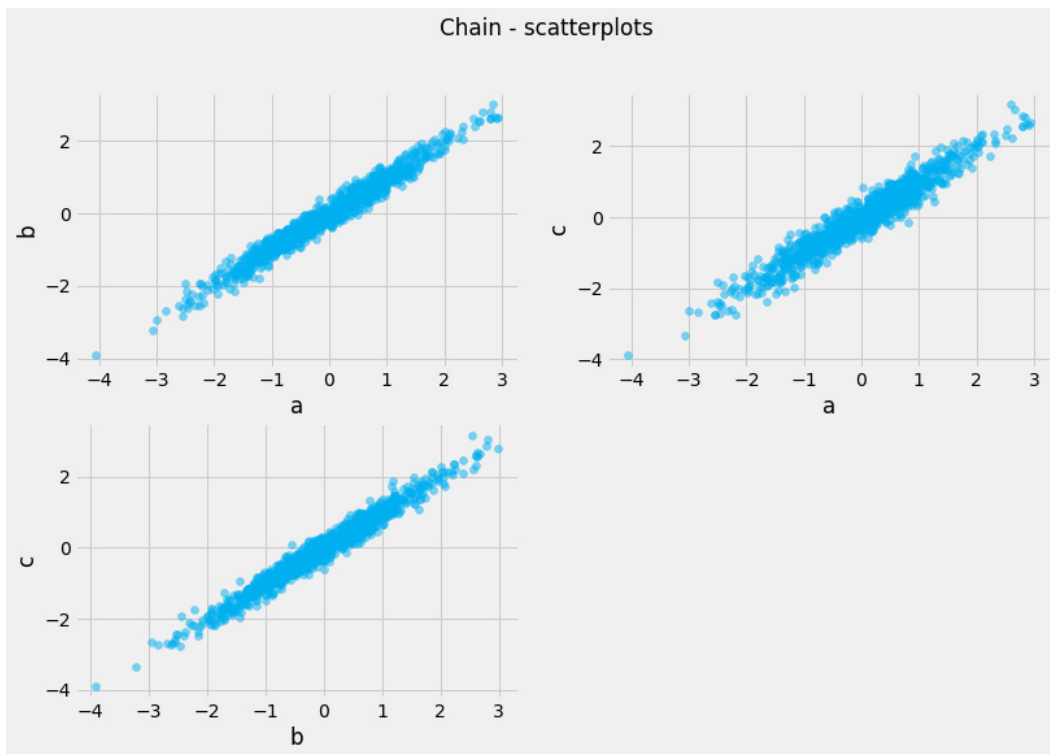
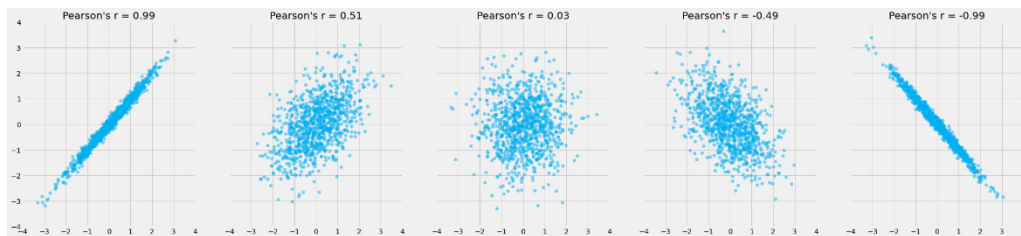


Figure 5.10 – Pairwise scatterplots for the dataset generated according to a chain-structured graph

We can see that all the scatterplots in *Figure 5.10* are very similar. The pattern is virtually identical for each pair of variables, and the correlation is consistently pretty strong. This reflects the characteristics of our data-generating process, which is linear and only slightly noisy.

How to read scatterplots

Scatterplots are a popular way to visualize bivariate data. Visual examination can help us quickly assess what kind of relationship (if any) exists between variables. A plot in this box shows us five different scatterplots with varying strengths of relationships between variables. In the leftmost panel, we see an almost perfect positive correlation (**Pearson's $r = 0.99$**), in the middle panel, there's virtually no relation between the variables. In the rightmost panel, there is an almost perfect *negative* correlation (**Pearson's $r = -0.99$**)



Note that the correlation metric that we used – **Pearson's r** – can only capture linear relationships between two variables. Metrics for non-linear relationships are also available, but we won't discuss them here.

Generating the fork dataset

Now, let's do the same for the fork structure. We'll start with generating the data:

```
b = np.random.randn(N_SAMPLES)
a = b + NOISE_LEVEL*np.random.randn(N_SAMPLES)
c = b + NOISE_LEVEL*np.random.randn(N_SAMPLES)
```

In *Figure 5.11*, we can see pairwise scatterplots for the fork. What's your impression? Do they look similar to the ones in *Figure 5.10*?

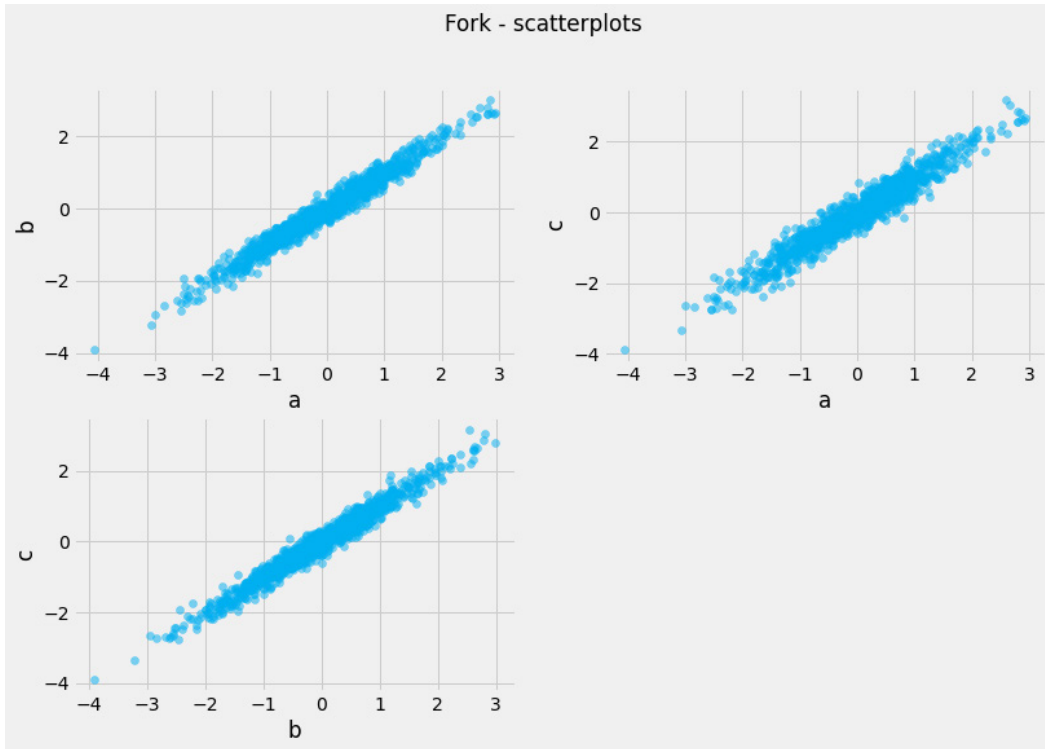


Figure 5.11 – Pairwise scatterplots for the dataset generated according to a fork-structured graph

Both figures (*Figure 5.10* and *Figure 5.11*) might differ in detail (note that we generate independent noise variables for each dataset), but the overall pattern seems very similar between the chain and fork datasets.

That's an interesting observation! What do you expect to see for colliders?

Let's see!

Generating the collider dataset

Let's start with the data:

```
a = np.random.randn(N_SAMPLES)
c = np.random.randn(N_SAMPLES)
b = a + c + NOISE_LEVEL*np.random.randn(N_SAMPLES)
```

Figure 5.12 presents pairwise scatterplots for the collider dataset:

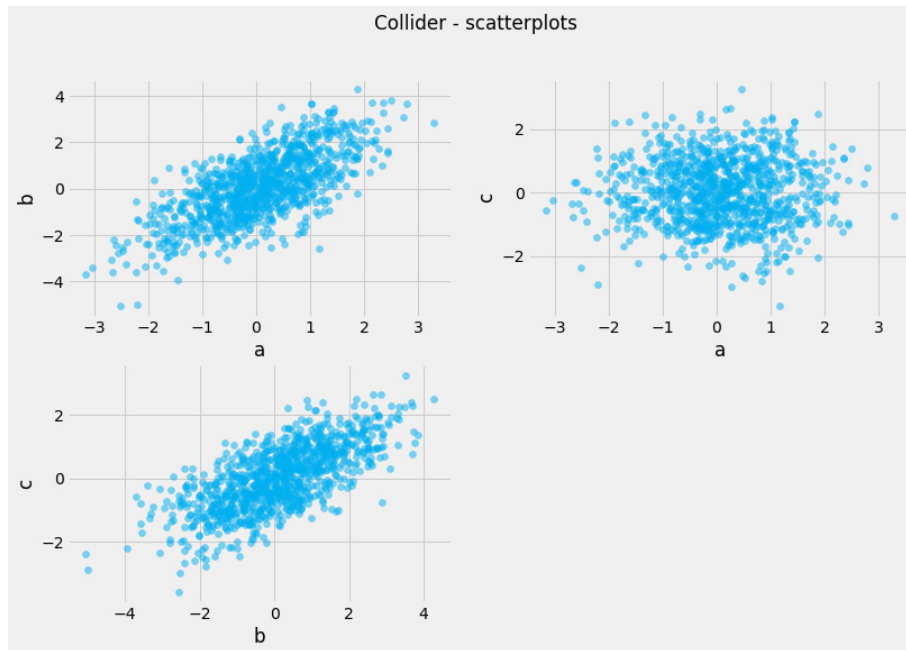


Figure 5.12 – Pairwise scatterplots for the dataset generated according to a collider-structured graph

This time the pattern is pretty different! Relationships between a and b (top left) and b and c (bottom left) seem to be noisier. Moreover, it seems there's no correlation between A and C (top right). Note that this result is congruent with what we said about the nature of colliders earlier in this chapter. Additionally, when we take a look at the data-generating process, this shouldn't be very surprising because the data generating process renders A and C entirely independent, evidence for, which you can also see in the code.

Fitting the regression models

Now, let's take all three datasets we generated and model them using multiple linear regression.

In each case, we'll regress C on A and B . We'll use `statsmodels` to fit the regression models. The code for each of the three models is identical:

```
X = pd.DataFrame(np.vstack([a, b]).T, columns=['A', 'B'])
X = sm.add_constant(X, prepend=True)
model = sm.OLS(c, X)
results = model.fit()
```

In the first line, we create a pandas dataframe using NumPy arrays containing our predictors (*A* and *B*). This will make `statsmodels` automatically assign the correct variable names in the model summary.

After fitting the models (for the full flow, check out the code in the notebook (<https://bit.ly/causal-ntbk-05>)), we're ready to print the model summaries and compare the results.

Figure 5.13 provides us with a compact summary of all three models. We used yellow and red ellipses to mark the *p*-values for each of the models:

	coef	std err	t	P> t	[0.025	0.975]
Chain 						
const	-0.0086	0.007	-1.320	0.187	-0.021	0.004
A	-0.0238	0.033	-0.729	0.466	-0.088	0.040
B	1.0217	0.032	31.645	0.000	0.958	1.085
Fork 						
const	0.0077	0.006	1.241	0.215	-0.004	0.020
A	0.0090	0.031	0.292	0.770	-0.052	0.070
B	0.9938	0.032	31.372	0.000	0.932	1.056
Collider 						
const	2.082e-17	1.35e-17	1.536	0.125	5.77e-18	4.74e-17
A	-1.0000	2.01e-17	-4.97e+16	0.000	-1.000	-1.000
B	1.0000	1.39e-17	7.19e+16	0.000	1.000	1.000

Figure 5.13 – The results of regression analysis of three basic conditional independence structures

In Figure 5.13, there are three rows printed out for each model. The top row is marked as `const`, and we'll ignore it. The remaining two rows are marked with `A` and `B`, which denote our variables, *A* and *B*, respectively.

For the sake of our analysis, we'll use the customary threshold of 0.05 for *p*-values. We'll say that *p*-values greater than 0.05 indicate a non-significant result (no influence above the noise level), and *p*-values lower than or equal to 0.05 indicate a significant result.

We can see that for the chain model, only one predictor (*B*) is significant.

Why is that?

We've seen a clear linear relationship between *A* and *C* in Figure 5.10. It turns out that these two observations are consistent. Multiple linear regression computes the effects of predictors on the expected value of the dependent variable, *given* all other variables.

It means that we no longer look at pairwise relationships (such as in the scatterplots earlier in this section) but rather at conditional pairwise relationships. The result of the regression analysis is also congruent with the intuition that we've built in the example at the beginning of this chapter.

When we look at the results for the fork, we see the same pattern.

Again, only one predictor is significant, and again it turns out to be *B*. The logic behind the difference between the pairwise scatterplot and regression result is virtually identical to the one that we just discussed for the chain.

The model for the collider gives us a different pattern. For this model, both *A* and *B* are significant predictors of *C*. If you now take a look at *Figure 5.12* again, you can clearly see that there's no relationship between *A* and *C*.

This non-existing relationship between *A* and *C* is the essence of **spurious relationships**! They are an artifact, but – interestingly – this artifact is not only a side-effect of the method we used, and you can observe them in real life!

Spurious relationships in real life

Many companies might hire people based on their skills and their personality traits. Imagine that company *X* quantifies a person's coding skills on a scale from one to five. They do the same for the candidate's ability to cooperate and hire everyone who gets a total score of at least seven. Assuming that coding skills and ability to cooperate are independent in the population (which doesn't have to be true in reality), you'll observe that in company *X*, people who are better coders are less likely to cooperate on average, and those who are more likely to cooperate have fewer coding skills. You could conclude that being non-cooperative is related to being a better coder, yet this conclusion would be incorrect in the general population. To see why this happens, recall the water glasses example from the *Colliders, immoralities, or v-structures* section and think of coding skills and cooperativeness as water. Controlling for the *hiring status* is like keeping the total amount of *water* (coding skills + cooperation skills) constant. Pouring some water from one glass (e.g., coding skills glass) into another (cooperativeness glass) makes them negatively correlated when controlling for the hiring status (the total amount of water).

Before we conclude this section, I want to ask you to recall our discussion on **statistical control** from *Chapter 3*, and think again about the question that we asked in one of the sections – should we always control for all the available variables? – knowing what we've learned in this chapter.

We will see how to methodologically address the *to-control-or-not-to-control* question in the next chapter.

In this section, we saw how the properties of chains, forks, and colliders manifest themselves in the realm of statistical analysis. We examined pairwise relationships between variables and compared them to conditional relationships that we observed as a result of multiple regression analysis. Finally, we deepened our understanding of confounding.

Wrapping it up

This chapter introduced us to the three basic conditional independence structures – chains, forks, and colliders (the latter also known as immoralities or v -structures). We studied the properties of these structures and demonstrated that colliders have unique properties that make constraint-based causal discovery possible. We discussed how to deal with cases when it's impossible to orient all the edges in a graph and introduced the concept of MECs. Finally, we got our hands dirty with coding the examples of all the structures and analyzed their statistical properties using multiple linear regression.

This chapter concludes the first, introductory part of this book. The next chapter starts on the other side, in the fascinating land of causal inference. We'll go beyond simple linear cases and see a whole new zoo of models.

Ready?

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